Repository Organization Report

M5 Retail Demand Forecasting — Problem Statement 11

Reorganization, path migration and full integrity audit

NPN AIA Hackathon — St. Joseph's College of Engineering

Executed 2026-08-14. 357 files relocated, 0 files lost, 0 protected bytes changed.

Outcome: *the repository was reorganized from 14 loosely-named top-level folders into a conventional structure, with every protected artefact verified byte-identical afterwards and the pipeline confirmed still functional by a live run.*

1. Original structure

Fourteen top-level directories with inconsistent naming conventions (`raw_dataset`, `EDA`, `analysis_output`, `Project_Approach`, `FINAL_APPROACH`, `ProblemStatement_Walkthrough`), documentation spread across five of them, 31 scripts in one flat folder, 71 experiment records and 36 artifacts intermixed, 50 report files unfiled, and the final forecast sitting alongside 26 validation prediction files.

```
raw_dataset/        processed_dataset/   EDA/            analysis_output/
Project_Approach/   FINAL_APPROACH/      ProblemStatement_Walkthrough/
pipeline/  scripts/  experiments/  models/  predictions/  artifacts/  reports/
end_to_end_approach.md   requirements.txt
```

2. Final structure

```
README.md  PROJECT_INDEX.md  requirements.txt  .gitignore
data/raw/            the 5 original CSVs — IMMUTABLE
data/processed/      sales_long_full.parquet + build/audit reports
pipeline/            reusable source package (13 modules)
scripts/             6 stage folders, chronological numbering preserved
experiments/registry/    71 JSON records
experiments/artifacts/   36 result tables and diagnostics
experiments/EXPERIMENT_LEDGER.md
models/champion/     the selected model + its forecast retrain
models/experiments/  10 other experimental models
predictions/final_forecast/  the deliverable
predictions/validation/      26 backtest prediction files
reports/             5 stage folders + charts/
docs/                5 numbered folders, chronological
```

3. Major organizational decisions

Decision	Reasoning
`data/raw/` holds **only** the 5 source CSVs	The dataset guides that lived inside `raw_dataset/Dataset_Explanation/` are documentation, not data. Moving them to `docs/02_dataset/` makes the raw folder purely immutable source.
Kept the package named `pipeline/` rather than renaming to `src/`	It is descriptive and already correct; renaming would have touched 31 import statements for no functional gain.
Experiment records kept **flat** in `registry/`	The 71 JSONs are a ledger, not 71 projects. Seventy-one folders each holding one file would be worse navigation. `EXPERIMENT_LEDGER.md` provides the index instead.

Decision	Reasoning
Artifacts kept flat in `experiments/artifacts/`	Scripts write here via `config.ARTIFACTS_DIR`; subfoldering would break write paths on re-run. An in-place README maps every file to the run that produced it.
Final forecast separated from validation predictions	The deliverable should never be confused with 26 backtest files.
Champion separated from other models	`models/champion/` is curated by hand; new training runs write to `models/experiments/`.
Reports filed by **stage**, md+pdf pairs together	Grouping by format would separate each report from its source.
Script bootstrap made depth-independent	Scripts now locate the project root by walking up to the folder containing `pipeline/config.py`, so they keep working wherever they are filed.

4. Files moved

357 files relocated. By destination:

Destination	Files
`experiments/registry/`	71
`docs/04_eda/tables/`	38
`experiments/artifacts/`	36
`docs/04_eda/charts/`	26
`predictions/validation/`	26
`docs/03_exploratory_analysis/`	24
`reports/04_optimization/`	22
`reports/02_modelling/`	16
`models/experiments/`	10
`docs/04_eda/statistics/`	9
`scripts/04_optimization/`	9
`docs/05_approach/`	7
`scripts/02_modelling/`	7
`reports/05_diagnostics_and_research/`	6
`scripts/03_benchmark_investigation/`	6
`data/raw/`	5
`docs/02_dataset/`	5
`data/processed/_audit/`	5

The complete move log — every source path and destination path — is preserved verbatim in `experiments/artifacts/repository_move_log.json`.

5. Files renamed

From	To	Why
`end_to_end_approach.md` (project root)	`docs/01_problem_statement/TEAM_end_to_end_approach.md`	It is the other team's document; the prefix prevents it being mistaken for ours, and it no longer clutters the root.

No other file was renamed. Experiment ids, model filenames and prediction filenames were left exactly as they were, so every reference inside the 71 JSON records and 25 reports still resolves by name.

6. Files removed

Removed	Count	Why it is safe
pipeline/__pycache__/	1 directory	Compiled bytecode, regenerated automatically on next import
Empty source directories	13	Left behind after their contents moved; contained nothing

Nothing else was deleted. In particular, four groups of byte-identical model and prediction files were found and **deliberately kept**: they are independent re-runs of the same configuration that reproduce identical output, which is the project's evidence that the pipeline is deterministic. Removing them as "duplicates" would have destroyed that evidence.

7. Protected file integrity — before vs after

File (final location)	MD5	Match
data/raw/calendar.csv	3ffeab2991b0c8e861d008b39ea4c95c	MATCH
docs/02_dataset/Cognizant_M5_Problem_Approach_Algorithms_Hierarchical_Plan.pdf	886d7de7ce7e725a48f014ecaaa3dbe0	MATCH
docs/02_dataset/Cognizant_Walmart_M5_Dataset_Deep_Explanation.pdf	9b29d38c7c593793579dbdf8e8099a7d	MATCH
docs/02_dataset/DATASET_EXPLAINED.md	f55aaab94fbe61d35a17d2a26404ba1b	MATCH
docs/02_dataset/DATASET_EXPLAINED.pdf	cebeea524f08cee7609f1e39a3248c3c	MATCH
docs/02_dataset/DATASET_SUMMARY.md	04d6961a5581bb3f04b3b4d670533a0b	MATCH
data/raw/sales_train_evaluation.csv	b806dfc9f30a745102b708c09951f6aa	MATCH
data/raw/sales_train_validation.csv	26a366a25beb57b0a8f4c7b148758f94	MATCH
data/raw/sample_submission.csv	c281a69d7c011274899d92020a66e25b	MATCH
data/raw/sell_prices.csv	08c591caa99e55daf3e0ccac913f7c85	MATCH
data/processed/sales_long_full.parquet	faa70ced1dddd2a84801a748b1149986	MATCH
predictions/final_forecast/final_forecast_28day.csv	d1067b9ff7a09fefadac71be27b7044c	MATCH
predictions/final_forecast/submission_m5_format.csv	1722effc629d8d74000c4c33c7ae36ca	MATCH
predictions/validation/model_04_tweedie_recency_listing_validation.csv	36553293c0d30e15b60f712a17dd4721	MATCH
models/champion/model_04_tweedie_recency_listing.txt	e1fb7b8733b0e030174fed3b2648f16a	MATCH
models/champion/model_07_final_forecast.txt	d722e2cc6ae11ae6f6d78ca7322a50e9	MATCH

All 16 protected files are byte-identical to their pre-reorganization state: True.

8. Integrity checks

#	Check	Result
1	Raw dataset hashes unchanged	PASS (5/5)
2	Final forecast file unchanged	PASS
3	Champion validation predictions unchanged	PASS

#	Check	Result
4	Champion RMSE	2.121043 — PASS
5	Champion MAE	1.031927 — PASS
6	All 71 experiments present and parsing	71/71 — PASS
7	All reports accessible and opening	25 PDFs — PASS
8	No broken internal paths	PASS — 32 scripts compile; live data load, feature build and champion prediction all succeeded
9	No duplicate final artefacts created	PASS — `final_forecast/` holds exactly 2 files, `champion/` exactly 2 models
10	Repository navigable from root	PASS — README.md + PROJECT_INDEX.md
11	Final model unambiguously identifiable	PASS — `models/champion/` with an in-place README explaining the duplicate experiment ids
12	No experiment modified by the cleanup	PASS — spot-checked metrics across Experiments #4, #69, #70, #71

Check 8 was verified functionally, not just by inspection: after the move, `M5Data()` loaded the raw CSVs, `FeatureBuilderV2` built a validation frame, and the champion model file loaded and produced predictions.

9. Ambiguous items left for review

Item	Where it is	Note
`docs/05_approach/_build_pdf.py`	with its document	A one-off renderer for `FINAL_PROJECT_APPROACH.md`, superseded by `pipeline/report_pdf.py`. Kept beside the document it builds because it resolves paths relative to its own location. Safe to delete if that document is never rebuilt.
`docs/03_exploratory_analysis/`	24 files	First-pass exploration that predates the formal EDA. Superseded in substance but referenced by `docs/02_dataset/DATASET_SUMMARY.md`, so retained.
`.claude/settings.local.json`	project root	Tooling configuration, left untouched.

Nothing was placed in a quarantine folder — every file had a clear home.

10. Entry points

Purpose	Path
Understand the project	`README.md`
Find any artefact	`PROJECT_INDEX.md`
Navigate the 71 experiments	`experiments/EXPERIMENT_LEDGER.md`
Verify integrity and leakage	`scripts/01_foundation/01_foundation_check.py`
Regenerate the forecast	`scripts/02_modelling/08_final_forecast.py`
The deliverable	`predictions/final_forecast/final_forecast_28day.csv`

Repository organization only. No model was retrained, no parameter changed, no prediction value altered, and no experiment conclusion revised.